

HORNSBY GIRLS' HIGH SCHOOL



MATHEMATICS YEAR 11 Preliminary Assessment Task 2 Half-Yearly Examination 2009

Student Number: _____

Time Allowed: 90 minutes plus 5 minutes reading time

Instructions:

- Attempt all questions
- Start a new page for each question
- The marks for each question are indicated
- Show all necessary working
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used

Outcomes Assessed:

- P1 Demonstrates confidence in obtaining realistic solutions to problems
- P2 Provides reasoning to support conclusions in the correct context
- P3 Performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4 Chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5 Understands the concept of a function and the relationship between a function and its graph

Marking Scheme:

	Total	
Algebra and Arithmetic	Q1.	/14
Functions and Graphs	Q2.	/14
Linear Functions	Q3.	/14
Trigonometric Functions	Q4.	/14
	Q5.	/14
Total Marks:		/70

This assessment task constitutes 30% of the final Preliminary Course Assessment.

QUESTION 1 (14 Marks) START A NEW PAGE**Marks**

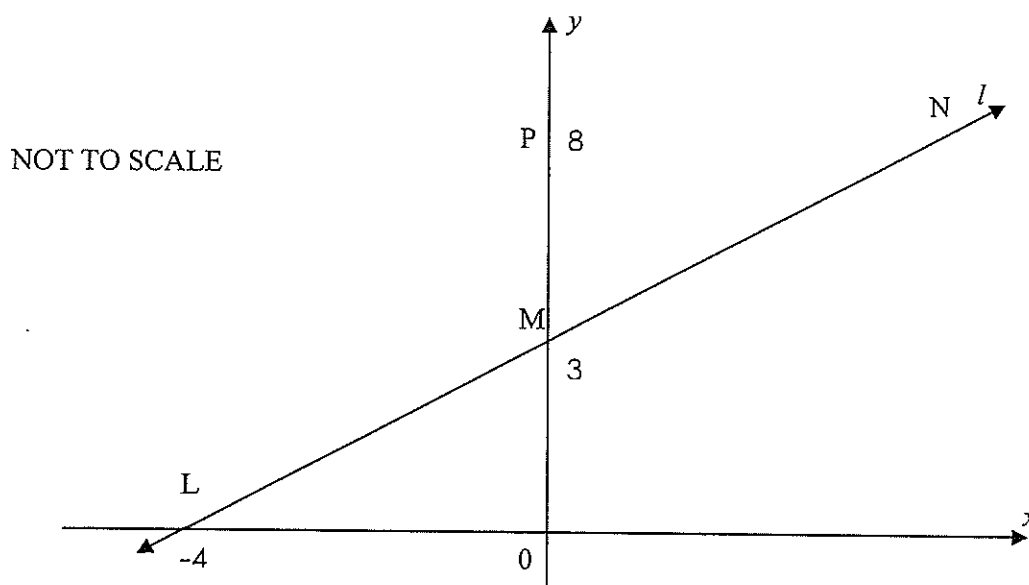
- a) Evaluate: $\sqrt{\frac{1.25 \times (3.6^4)}{0.075}}$ correct to 3 significant figures. 1
- b) Factorise: $3y^3 - 81$ 2
- c) Simplify: $(\sqrt{17} - 2\sqrt{5})(\sqrt{17} + 2\sqrt{5})$ 2
- d) Show that the only real solution of $|4x - 1| = 3 - 5x$ is $x = \frac{4}{9}$ 3
- e) If $f(x) = x^2 - 8x$, find $\frac{f(a+h) - f(a)}{h}$ 3
- f) Solve for x : $|7 - 2x| > 9$ and graph the solution on a number line. 3
-

QUESTION 2 (14 Marks) START A NEW PAGE**Marks**

- a) State the domain and range for (i) $y = \frac{8}{x-1}$ 2
- (ii) $y = \sqrt{16 - x^2}$ 2
- b) If $f(x) = 2^{-x} + 2^x$,
(i) find $f(a)$ and $f(-a)$. 1
(ii) hence, determine whether the function $f(x)$ is odd, even or neither. 1
- c) Sketch the following functions on separate number planes showing important features and intercepts. 8
- (i) $y = (x + 2)^2$
- (ii) $y = |x + 2|$
- (iii) $y = \frac{1}{x+2}$
- (iv) $y = 2 - x^3$

QUESTION 3**(14 Marks)****START A NEW PAGE****Marks**

a)



The line l cuts the x axis at $L (-4,0)$ and the y axis at $M (0,3)$. N is a point on the line l and P is the point $(0,8)$

- | | | |
|-------|--|---|
| (i) | Find the equation of the line l in general form. | 2 |
| (ii) | Show that $\triangle LMP$ is isosceles. | 2 |
| (iii) | M is the midpoint of LN find the coordinates of N . | 2 |
| (iv) | Show that $\angle NPL$ is a right angle. | 2 |
| (v) | Find the perpendicular distance from the point $Q(-4,5)$ to the line l . | 2 |
| (vi) | Find the gradient of PQ and hence determine if PQ is parallel to line l , LN . | 2 |

- b) Find the equation of the straight line that makes an angle of 135° with the positive direction of x axis and passes through the point $(2,-3)$. 2

QUESTION 4

(14 Marks)

START A NEW PAGE

Marks

a) Simplify (i) $\frac{1}{\sec(-\theta)} = \frac{1}{\sec \theta} = \cos \theta$

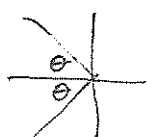
(ii) $\frac{\csc(180^\circ - \theta) \cdot \sin(90^\circ - \theta)}{\csc \theta - \cos \theta} = \cot \theta$

(iii) $2 - 2\sin^2 \theta = 2(1 - \sin^2 \theta) = 2\cos^2 \theta$

1
2
1
4

b) Solve $\cos \theta = -0.6$ for $0^\circ \leq \theta \leq 360^\circ$. Give your answer to the nearest minute.

2
5



$\theta = 126^\circ 52'$
 $233^\circ 8'$

c) Evaluate, giving the exact value as a single fraction for

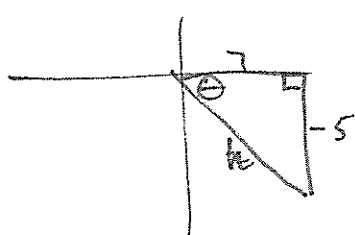
(i) $\tan 210^\circ = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}$

(ii) $\sin 60^\circ + \tan 210^\circ = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} = \frac{5\sqrt{3}}{6}$

1
2

d) Given that $\tan \theta = \frac{-5}{7}$ and $\cos \theta > 0$, find the exact value of $\csc \theta$. Show necessary working.

3
6



$h^2 = 49 + 25$
 $= 74$
 $h = \sqrt{74}$

$\therefore \csc \theta = \frac{1}{\sin \theta} = \frac{\sqrt{74}}{-5}$

e) Prove that $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$

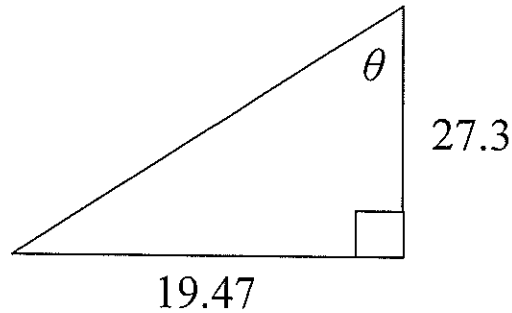
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LHS = $\frac{1 - \tan^2 \theta}{\sec^2 \theta}$
 $= \frac{1 - \frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{1} = \cos^2 \theta - \sin^2 \theta = \text{RHS}$

QUESTION 5 (14 Marks)**START A NEW PAGE****Marks**

- a) Find θ to the nearest minute.

2



- b) A ship leaves port and sails 58 kilometres on a bearing of 138° . How far due south of the port is the ship? Draw a diagram and give your answer correct to the nearest kilometre.

2

- c) From an aircraft flying at 9 000m above sea level the angle of depression of a ship is $33^\circ 25'$. How far is the ship from the aircraft in a straight line? (Give your answer correct to nearest metre).

2

- d) Solve for $0^\circ \leq \theta \leq 360^\circ$

(i) $\operatorname{cosec} \theta = 2$

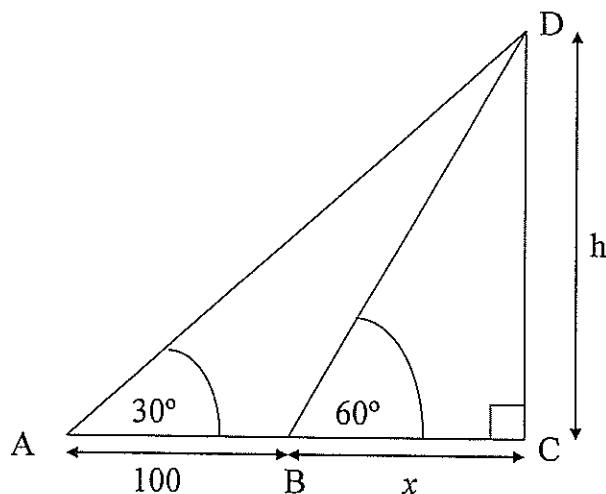
2

(ii) $\cot \theta = 0$

2

- e) A child at A observes the angle of elevation of a kite caught in a tree to be 30 degrees. She then walks 100 metres towards the kite to a point B and finds the angle of elevation to be 60 degrees. If the kite has a height of h metres and the child has x metres still to walk before she is directly under the kite, find x .

4

**END OF EXAMINATION**

Q1 a) $\div 52.9089 \dots$ (from calculator)
 $= 52.9$ (3 significant figures) | 1

b) $3(y^3 - 27) = 3(y-3)(y^2 + 3y + 9)$ | 2

c) $17 - 20 = -3$ | 2

d) $4x - 1 = 3 - 5x$ OR $-(4x - 1) = 3 - 5x$
 either $9x = 4$ $-4x + 1 = 3 - 5x$
 $x = \frac{4}{9}$ $x = 2$

Check: $|4 \times \frac{4}{9} - 1|$
 $LHS = \frac{7}{9}$

$RHS = 3 - 5 \times \frac{4}{9}$
 $= \frac{7}{9}$

$LHS = RHS$
 $\therefore x = \frac{4}{9}$ is the only solution.

Check:

$LHS = |4 \times 2 - 1|$
 $= 7$

$RHS = 3 - 5 \times 2$
 $= -7$

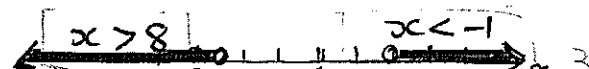
$\therefore$ Not a solution
 $LHS \neq RHS$

3

e) $f(a+h) = (a+h)^2 - 8(a+h)$
 $= a^2 + 2ah + h^2 - 8a - 8h$

$f(a) = a^2 - 8a$

$\therefore \frac{f(a+h) - f(a)}{h} = \frac{a^2 + 2ah + h^2 - 8a - 8h - (a^2 - 8a)}{h}$
 $= \frac{2ah + h^2 - 8h}{h}$
 $= \frac{h(2a + h - 8)}{h}$
 $= 2a + h - 8$ | 3

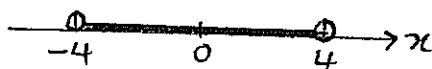
f) $|7 - 2x| > 9$
 either $-(7 - 2x) > 9$ OR $7 - 2x > 9$
 $7 - 2x < -9$ OR $-2x > 2$
 $-2x < -16$ $-2x > 2$
 $x > 8$ $x < -1$


Q2. a) i) D: $x \neq 1$

R: $y \neq 0$

(ii) D: $16 - x^2 \geq 0$

$(4-x)(4+x) \geq 0$



Test $x = 0$ $16 \geq 0$ ✓

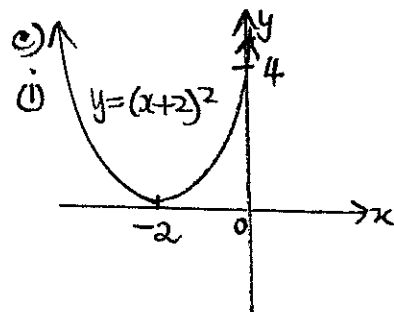
$\therefore -4 \leq x \leq 4$

R: $y \geq 0$

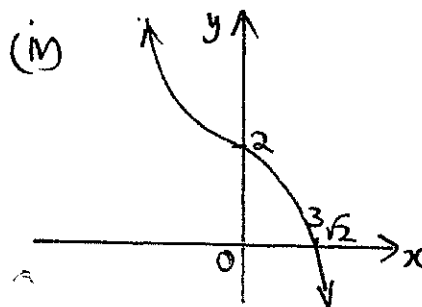
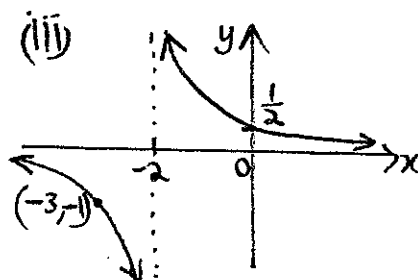
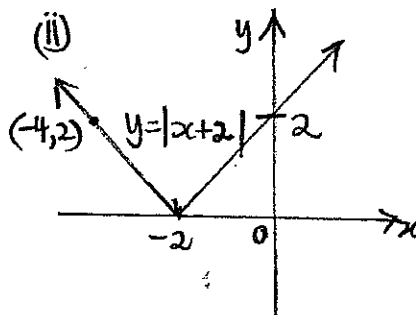
4

Q2 b) $f(a) = 2^{-a} + 2^a$
 $f(-a) = 2^a + 2^{-a}$

$\therefore f(-a) = f(a) \therefore$ Function is even. | 2



2 each $\left\{ \begin{array}{l} \text{shape} \\ \text{intercepts} \end{array} \right.$



2

Q3. a) i) $\frac{x}{-4} + \frac{y}{3} = 1$

$3x - 4y = 12$

$3x - 4y + 12 = 0$

ii) PM = 5 units

LM = 5 units {3, 4, 5} Pythagorean Triad

or $d = \sqrt{3^2 + 4^2}$
 $= \sqrt{25}$
 $= 5$

$\therefore \triangle LMP$ is isosceles (2 equal sides)

iii) $(0, 3) = \left(\frac{-4+x_1}{2}, \frac{0+y_1}{2} \right)$

$-4+x_1 = 0$, $\frac{0+y_1}{2} = 3$

$x_1 = 4$, $y_1 = 6$

$\therefore N$ is $(4, 6)$

iv) $m_{NP} = \frac{8-6}{0-4}$, $m_{PL} = \frac{8-0}{0+4}$
 $= -\frac{1}{2}$, $= 2$

$\therefore m_{NP} \times m_{PL} = -\frac{1}{2} \times 2$
 $= -1$

$\therefore NP \perp PL$

$\therefore \angle NPL$ is a right angle.

v) $d = \frac{|3(-4) - 4(5) + 12|}{\sqrt{3^2 + 4^2}}$

$= \frac{|-20|}{5}$ units.

vi) $m_{PQ} = \frac{8-5}{0+4}$, $m_{LN} = \frac{3}{4}$
 $= \frac{3}{4}$

$\therefore$ gradients are equal

$\therefore PQ \parallel LN$ (line l)

b) $m = \tan 135^\circ$
 $= -1$

$y = -1x + b$

$-3 = -1 \times 2 + b$

$b = -1$

$\therefore y = -x - 1$

Q4. a) i) $\cos(-\theta) = \cos \theta$

ii) $\frac{1}{\sin(180^\circ - \theta)} \times \frac{\cos \theta}{\sin \theta} = \frac{\cos \theta}{\sin \theta}$
 $= \cot \theta$

iii) $2(1 - \sin^2 \theta) = 2 \cos^2 \theta$

b) Quadrants II, III $\cos \theta < 0$

Acute angle α is $53^\circ 7' 48''$ (from calculator)

$\therefore \theta = 180^\circ - \alpha$ or $\theta = 180^\circ + \alpha$

$\div 126^\circ 52' 12''$ $\div 233^\circ 7' 48''$

$= 126^\circ 52'$ (correct nearest minute) $= 233^\circ 8'$ (correct nearest minute)

c) i) $\tan 210^\circ = \tan(180^\circ + 30^\circ)$ 3rd quadrant $\tan \theta > 0$
 $= \tan 30^\circ$
 $= \frac{1}{\sqrt{3}}$

ii) $\sin 60^\circ + \tan 210^\circ = \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}}$

$= \frac{3+2}{2\sqrt{3}}$

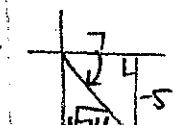
$= \frac{5}{2\sqrt{3}}$ or $\frac{5\sqrt{3}}{6}$

d) $\left. \begin{matrix} \tan \theta < 0 \\ \cos \theta > 0 \end{matrix} \right\} \therefore$ 4th quadrant.
 $\therefore \sin \theta < 0$

$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

$= \frac{1}{-\frac{5}{\sqrt{74}}}$

$= -\frac{\sqrt{74}}{5}$



(By Pythagoras' Thm)

$h = \sqrt{5^2 + 7^2}$
 $= \sqrt{74}$

e) LHS = $\left(1 - \frac{\sin^2 \theta}{\cos^2 \theta} \right) \div \sec^2 \theta$

$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\sec^2 \theta}$

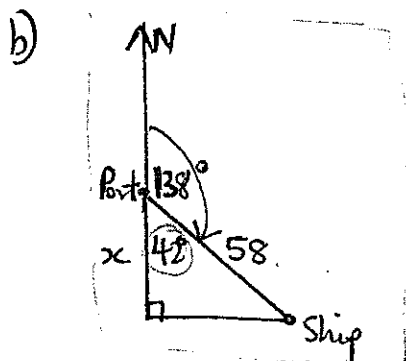
$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \times \cos^2 \theta$

$= \text{RHS.}$

$$Q5. a) \tan \theta = \frac{19.47}{27.3}$$

$$\theta \doteq 35^\circ 29' 45'' \text{ (from calc.)}$$

$$= \underline{35^\circ 30'} \text{ (to nearest minute)}$$

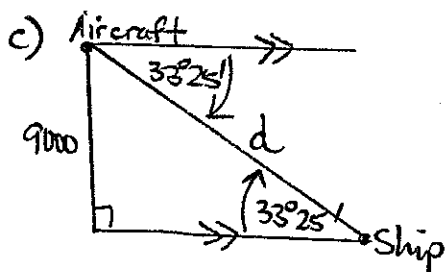


$$\cos 42^\circ = \frac{x}{58}$$

$$x = 58 \cos 42^\circ$$

$$\doteq 43.10 \dots \text{ (from calculator)}$$

$\therefore$ Ship is 43 km south of the port



$$\frac{9000}{d} = \sin 33^\circ 25'$$

$$d = \frac{9000}{\sin 33^\circ 25'}$$

$$\doteq 16342.13 \dots \text{ (from calculator)}$$

$\therefore$ The distance between them is 16 342 m.

d) i) $\operatorname{cosec} \theta = 2, 0^\circ \leq \theta \leq 360^\circ$

$$\therefore \frac{1}{\sin \theta} = 2$$

$$\sin \theta = \frac{1}{2} \quad \sin \theta > 0 \quad \therefore \text{I, II quadrants}$$

$$\theta = \underline{30^\circ}, \underline{180^\circ - 30^\circ}$$

$$= \underline{30^\circ}, \underline{150^\circ}$$

Q5d) ii) $\cot \theta = 0 \quad 0^\circ \leq \theta \leq 360^\circ$

$$\frac{1}{\tan \theta} = 0$$

$\tan \theta$ is undefined.

$$\therefore \theta = \underline{90^\circ}, \underline{270^\circ}$$

e)

$$\frac{h}{x} = \tan 60^\circ$$

$$h = x \tan 60^\circ$$

$$\frac{h}{(100+x)} = \tan 30^\circ$$

$$h = (100+x) \tan 30^\circ$$

$$x \tan 60^\circ = 100 \tan 30^\circ + x \tan 30^\circ$$

$$x \tan 60^\circ - x \tan 30^\circ = 100 \tan 30^\circ$$

$$x (\tan 60^\circ - \tan 30^\circ) = 100 \tan 30^\circ$$

$$x = \frac{100 \tan 30^\circ}{\tan 60^\circ - \tan 30^\circ}$$

$$= \frac{100 \times \frac{1}{\sqrt{3}}}{\sqrt{3} - \frac{1}{\sqrt{3}}}$$

$$= \frac{100}{\sqrt{3}} \div \left(\frac{3-1}{\sqrt{3}} \right)$$

$$= \frac{100}{\sqrt{3}} \times \frac{\sqrt{3}}{2}$$

$$= \underline{50}$$

